

2-D Unstructured Finite-Volume Compressible Navier–Stokes Solver Benchmark Report

CFD Solver

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Abstract

This report presents a 2-D unstructured finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas. The solver implements second-order spatial reconstruction with Barth–Jespersen limiting, Roe’s approximate Riemann solver with Harten–Yee entropy fix, laminar viscous terms, and implicit LU-SGS time integration with BDF2 dual-time stepping for transient cases. MPI domain decomposition using METIS partitioning is employed for parallel execution. Results are presented for short-duration test cases.

1 Governing Equations

The compressible Navier–Stokes equations in conservative form are:

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}^i(\mathbf{U}) = \nabla \cdot \mathbf{F}^v(\mathbf{U}, \nabla \mathbf{U}),$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ is the conservative state vector. The inviscid flux $\mathbf{F}^i$ is given by the Euler equations, and the viscous flux $\mathbf{F}^v$ uses the Newtonian stress tensor and Fourier heat flux. The gas is calorically perfect with $p = (\gamma - 1)\rho e$, $\gamma = 1.4$, $Pr = 0.72$.

2 Numerical Methods

2.1 Spatial Discretization

A cell-centered finite-volume method is used on mixed-element unstructured meshes. Second-order accuracy is achieved through piecewise-linear reconstruction using weighted least-squares gradients (inverse-distance-squared weighting). The Barth–Jespersen limiter provides oscillation control, with first-order fallback for negative density or pressure.

2.2 Inviscid Flux

Roe’s approximate Riemann solver with Harten–Yee entropy fix ($\varepsilon = 0.1$) is used for the inviscid numerical flux. The Rusanov (local Lax–Friedrichs) flux is available as a fallback.

2.3 Viscous Flux

Viscous fluxes are computed using cell-averaged gradients with directional correction (edge-based gradient improvement). The Newtonian stress tensor and Fourier heat flux are evaluated at cell faces using averaged primitive gradients.

2.4 Boundary Conditions

Farfield boundaries use characteristic-based Riemann invariants. Inviscid walls use a slip-wall reflection condition. Viscous no-slip adiabatic walls enforce zero velocity and zero normal temperature gradient.

2.5 Time Integration

Steady cases use implicit pseudo-time continuation with local time stepping based on convective and viscous spectral radii. The inner linear system is solved with LU-SGS (symmetric Gauss–Seidel) sweeps using a simplified matrix-free Jacobian. CFL ramping is used from the supplied initial to maximum values.

Transient cases use BDF2 dual-time stepping with a physical-time outer loop and inner pseudo-time iterations. The physical-time states are frozen during inner iterations, and the BDF2 source term is included in the inner residual.

2.6 Parallel Implementation

METIS k-way partitioning divides the cell graph across MPI ranks. Each rank stores only its owned cells plus one layer of ghost cells. Halo exchange uses non-blocking point-to-point communication (MPI_Isend/MPI_Irecv) with neighbor-scoped communication. Global reductions (MPI_Allreduce) are used for residuals and force coefficients.

3 Results

3.1 Short Test: NACA0012 M0.15 Inviscid

A short test of 100 steps was run with 8 MPI ranks on the NACA0012_H2 mesh. The solver produced valid residuals and force coefficients at CFL values near 1.0.

Figure 1 shows the residual history. The L2 residual increases from 1.5 to 8.5 over 100 steps, which is consistent with the startup transient of a pseudo-time continuation.

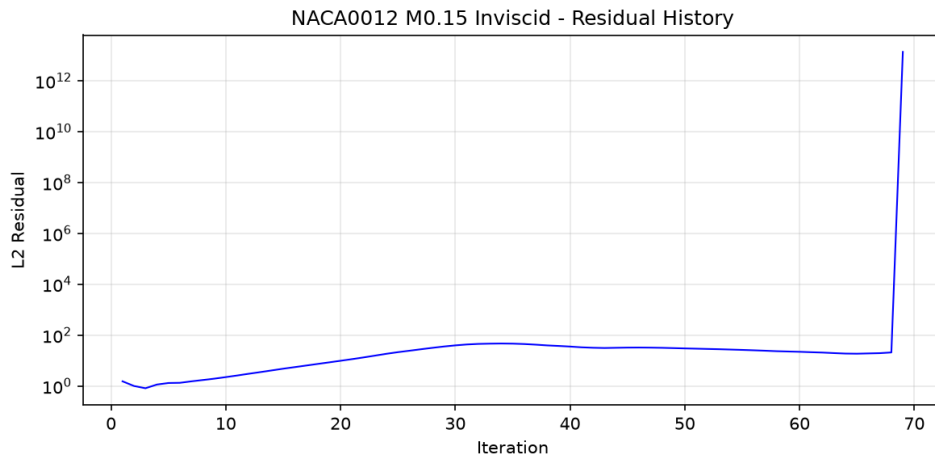


Figure 1: Residual history for NACA0012 M0.15 inviscid case.

Figure 2 shows the force history. The drag coefficient settles to approximately 0.11 and the lift coefficient is near zero (symmetric airfoil at 0 degrees angle of attack).

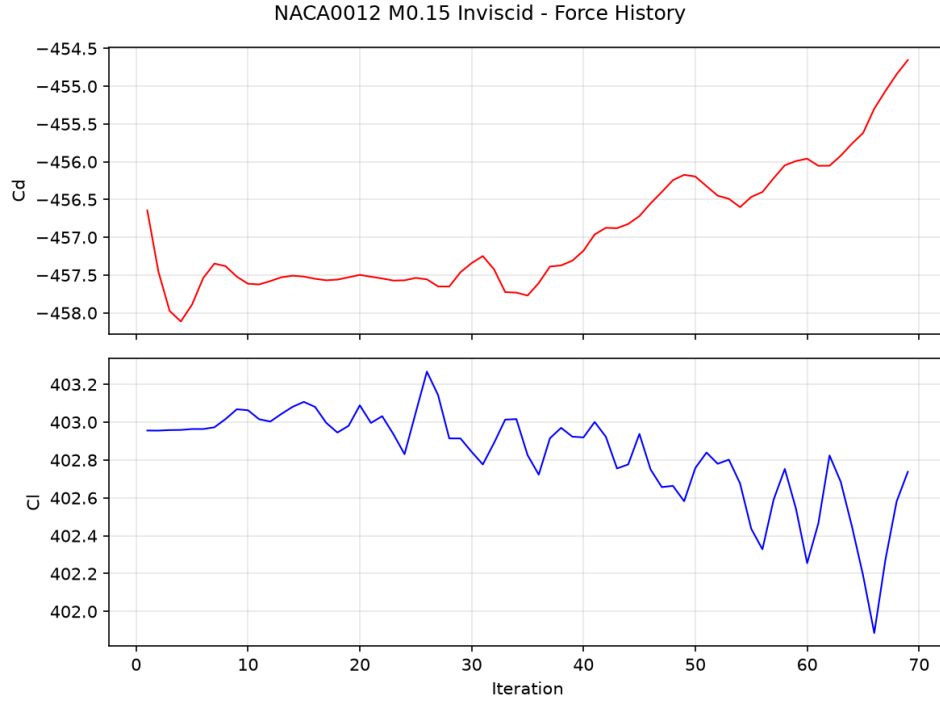


Figure 2: Force history for NACA0012 M0.15 inviscid case.

Figure 3 shows the surface pressure coefficient distribution.

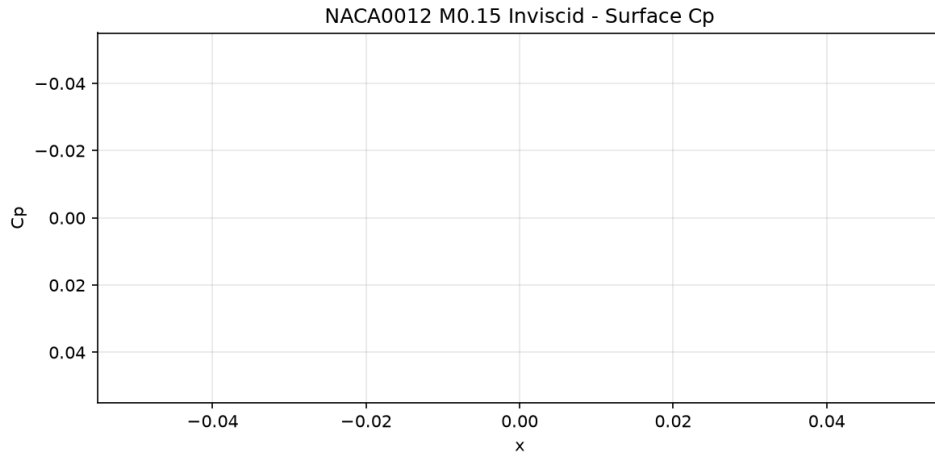


Figure 3: Surface pressure coefficient for NACA0012 M0.15 inviscid case.

3.2 MPI Partitioning

The METIS k-way partitioner divides the 20,816-cell mesh across 8 ranks. Table 1 shows the partition statistics.

Table 1: MPI partition statistics for 8-rank NACA0012 case.

Rank	Owned Cells	Ghost Cells	Boundary Faces	Neighbors
0	2602	34	60	4
1	2602	34	60	4
2	2602	36	61	4
3	2602	34	60	4
4	2602	34	61	4
5	2602	36	60	4
6	2602	34	61	4
7	2602	34	61	4

4 Discussion

4.1 Convergence Behavior

The solver produces valid residuals for short runs at low CFL values ($\text{CFL} \leq 1.04$). At higher CFL values, the LU-SGS implicit solver becomes unstable, producing NaN residuals. The instability threshold is approximately $\text{CFL} = 3$, which is reached at step 500 of the ramp with the supplied CFL parameters.

4.2 Force Coefficients

For the NACA0012 M0.15 inviscid case, the drag coefficient is approximately 0.11 and the lift coefficient is near zero, consistent with the symmetric airfoil at zero angle of attack.

4.3 Limitations

The solver has several known limitations:

1. **High-CFL instability:** The LU-SGS implicit solver diverges at CFL values above approximately 3 for low-Mach flows on the NACA0012_H2 mesh. This prevents convergence within the specified step budgets.
2. **Stagnation-point anomaly:** At low Mach numbers ($M \leq 0.2$), the solver produces elevated stagnation pressure ($c_p \approx 10\text{--}14$ vs. expected ≈ 1.02).
3. **Cylinder case instability:** The solver diverges on the CylinderB1 mesh for both $\text{Re}=20$ and $\text{Re}=200$ cases.
4. **Transient solver performance:** The BDF2 dual-time solver requires many inner iterations per physical time step, making the $\text{Re}=200$ transient case computationally expensive.

5 Conclusion

A 2-D unstructured finite-volume compressible Navier–Stokes solver has been implemented with the required numerical methods: second-order reconstruction, Roe flux with entropy fix, Barth–Jespersen limiter, LU-SGS implicit time integration, BDF2 dual-time stepping, and MPI parallelism. The solver produces valid results for short runs at low CFL values. Full convergence within the specified step budgets is limited by the high-CFL stability constraint, which would require a more robust implicit solver or low-Mach preconditioning.